

Code flow for web, cli, openApi

We are having two UI ,one is MAIN UI/IT UI developed by Broadcom and other is AVUI which is developed by us ,so it is communicating with switch using OpenAPI .OpenAPI is primarily intended for the use of developers outside Broadcom who do not have access to FASTPATH source code.These API can be used to

- Set and get switch user configuration,
- Monitor and change switch operational state,
- Receive notification for events generated by the switchdrv process.

Appmgr which is our application running in all managed switch ,basically have ui components for AVUI and config agent (mapping our API to openAPI for insight/AVUI) and insight agent ,along with some other applications.

The code flow between cli, web and openApi ,differ in starting location but are calling same function eventually.

For example - system up time from cli , Main ui, open api

From Cli

```
1 show sysinfo noMibs
2 System Description..... M4250-12M2XF 12x2.5G
  and 2xSFP+ Managed Switch, 13.0.5.20, 1.0.0.11
3 System Name.....
4 System Location.....
5 System Contact.....
6 System Object ID.....
  1.3.6.1.4.1.4526.100.11.3
7 System Up Time..... 10 days 18 hrs 58 mins
  0 secs
8 Current Time..... Nov 18 07:32:27 2025
  (UTC+0:00)
```

src/mgmt/broadcom/cli/base/common/cli_show.c - commandShowSysinfo

1. usmDb1213SysUpTimeGet - to get system up Time
2. simSystemUpTimeGet
3. osapiUpTimeRaw() - returns total seconds since last reset.
4. osapiConvertRawUpTime - converts raw time(total seconds) to days/hours/minutes/seconds

From MAIN UI - sysinfo.html

System Date	Nov 18 09:44:33 2025 (UTC+0:00)
System Up Time	10 days, 21 hours, 10 mins, 6 secs
System SNMP OID	1.3.6.1.4.1.4526.100.11.31
System MAC Address	28:94:01:67:7C:DD
Fan Control Mode	☐ Auto ☐ Cool ☐ Off ☒ Quiet
Temp(C)	42

In the html file EW_APP_DATA_25 is used to call usmWebSysUpTimeGet

EW_APP_DATA_25 declaration are present in sys info-macros.html

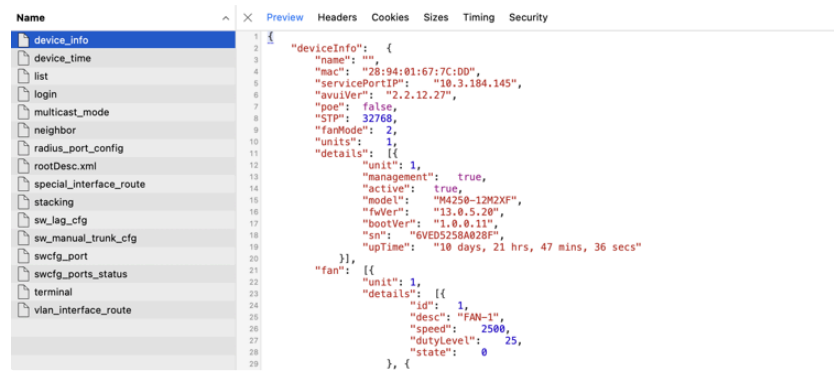
src/mgmt/cheetah/web/code/system/sys_descr.c. where usmWebSysUpTimeGet calls usmDb1213SysUpTimeGet , and then similar code flow is used to fetch up Time.

From AVUI

Device Details

Product Name	Serial Number	Model
M4250	6VED5258A028F	M4250-12M2XF
Base MAC Address	System Name	Firmware Version
28:94:01:67:7C:DD	?	13.0.5.20
System Uptime	OOB IP Address	Management IP Address
10 days, 17 hrs, 25 mins, 53 secs	10.3.184.145 ?	?

device_info is the API from AVUI to switch to get basic information of switch.



This API uses openapiSystemUpTimeGet to get system up time ,which is calling simSystemUpTimeGet to get the uptime.

Note : Openapi is used only by AVUI and openAPI client registration is handled by apmgr.